

Second Welfare Theorem (SWT)

Preliminaries SWT

- Let $B \subset \mathbb{R}$, $a \in \mathbb{R}$, then
 - $\sup B \leq a \implies [(\forall b \in B) b \leq a]$
 - $\inf B \geq a \implies [(\forall b \in B) b \geq a]$
- Let A_1, A_2 be subsets of a vector space X . Then,
- $A_1 + A_2 = \{x_1 + x_2 : x_1 \in A_1, x_2 \in A_2\}$.
- $\text{co } A_1$ is the convex hull of A_1
- If A_1, A_2 are convex, then $A_1 + A_2$ is convex.
- $p \cdot A_1 = \{p \cdot x_1 : x_1 \in A_1\}$.

Separating Hyperplane Theorem

Theorem 1 (Minkowski) If $B, C \subseteq \mathbb{R}^L$ are convex and $B \cap C = \emptyset$, then $\exists p \in \mathbb{R}^L, p \neq 0$, such that $\sup p \cdot B \leq \inf p \cdot C$.

In particular, $b \in B, c \in C \implies p \cdot b \leq p \cdot c$.

Second Welfare Theorem

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Weak second welfare theorem

Theorem 2 (WSWT) Let $\mathcal{E}$ be an Arrow-Debreu economy with convex production sets and complete, transitive, convex, LNS preferences. Let (x^*, y^*) be a Pareto optimal allocation. Then $(\exists p^* \in \mathbb{R}^L, p^* \neq 0)$ such that

1. $(\forall j) (\forall y_j \in Y_j) \quad p^* \cdot y_j^* \geq p^* \cdot y_j$ (price-taking profit maximization)
2. $(\forall i) [x_i \succsim_i x_i^* \implies p^* \cdot x_i \geq p^* \cdot x_i^*]$ (price-taking "quasi preference maximization")
3. $\sum_{i=1}^I x_i^* = \bar{\omega} + \sum_{j=1}^J y_j^*$

Any (x^*, y^*, p^*) satisfying 1-3 is a quasi-equilibrium.

Proof

Since (x^*, y^*) is an allocation, (3) holds. We will prove (1) and (2) in several steps. Let

$$V_i = \{x_i \in X_i : x_i \succ_i x_i^*\}$$

$$V = \sum_{i=1}^I V_i$$

$$Y = \sum_{j=1}^J Y_j$$

Claim (i). V , Y , and $\{V_i\}_{i=1}^I$ are convex. It follows from convexity of preferences and production sets and because the sum of convex sets is a convex set.

Claim ii. $V \cap (\bar{\omega} + Y) = \emptyset$

- Suppose not. Then $\exists \sum_{i=1}^I x_i \in V : \sum_{i=1}^I x_i \in (\bar{\omega} + Y)$.
- Hence (x^*, y^*) is not PO because for every i , $x_i \succ_i x_i^*$, and there is $y \in Y$ such that (x, y) is a feasible allocation.

This contradiction proves the claim.

Claim iii. $\exists p^* \in \mathbb{R}^L, p^* \neq 0 : \inf p^* \cdot \bar{V} \geq \sup p^* \cdot (\bar{\omega} + Y)$

- V and $\bar{\omega} + Y$ are nonempty, convex and do not intersect. By SHT,

$\exists p^* \in \mathbb{R}^L, p^* \neq 0 : \inf p^* \cdot V \geq \sup p^* \cdot (\bar{\omega} + Y)$. Thus

$$\left(\sum_{i=1}^I x_i \in V \right) \wedge \left(\sum_{j=1}^J y_j \in Y \right) \implies \left[p^* \cdot \sum_{i=1}^I x_i \geq p^* \cdot \left(\bar{\omega} + \sum_{j=1}^J y_j \right) \right]$$

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- If $z \in \overline{V}$, then $\exists z^n \in V : z^n \rightarrow z$ (definition of closure)
 - $p^* \cdot z^n \geq \sup p^* \cdot (\bar{\omega} + Y)$
 - Continuity of dot product implies $p^* \cdot z \geq \sup p^* \cdot (\bar{\omega} + Y)$.

Recall for future use that $\bar{\omega} + Y \ni \bar{\omega} + \sum_j y_j^* = \sum_i x_i^*$.

Claim iv. $\forall z \in \overline{V}, \forall y \in Y,$

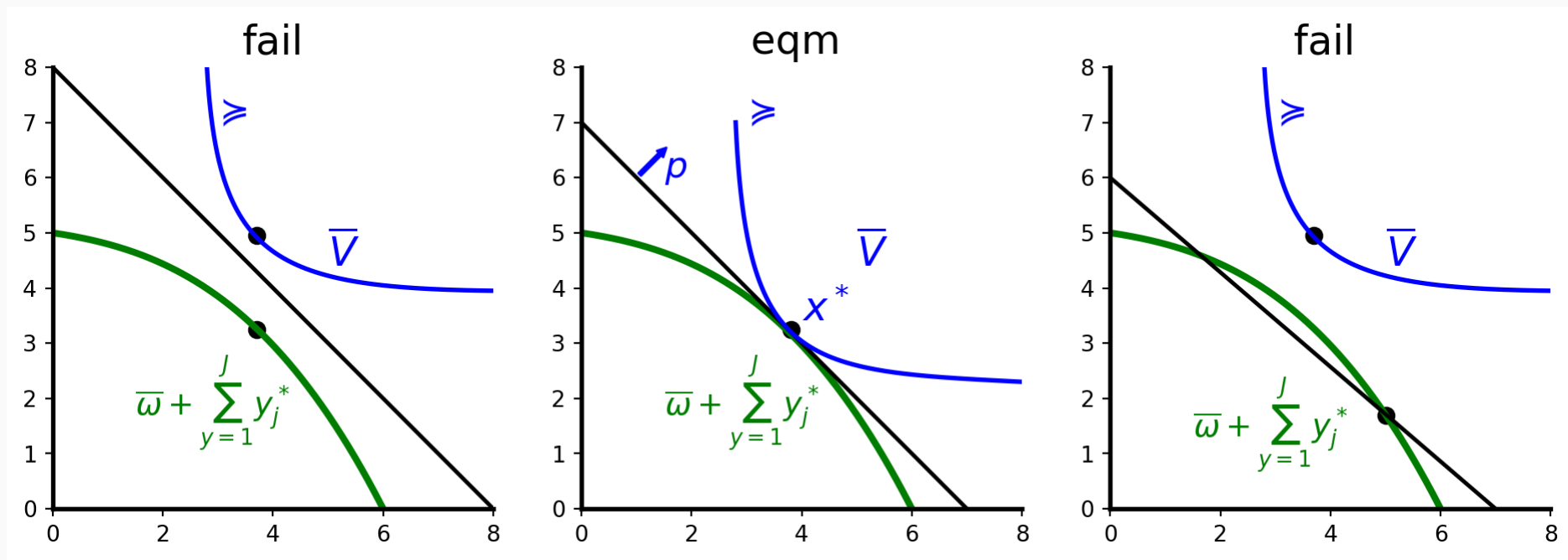
$$p^* \cdot z \geq p^* \cdot \sum_{i=1}^I x_i^* = p^* \cdot (\bar{\omega} + \sum_{j=1}^J y_j^*) \geq p^* \cdot (\bar{\omega} + \sum_{j=1}^J y_j)$$

Proof. By LNS for $i = 1, \dots, I,$

$$\forall \epsilon > 0, \exists \tilde{x}_i^\epsilon, \|\tilde{x}_i^\epsilon - x_i^*\| < \epsilon \wedge \tilde{x}_i^\epsilon \in V_i$$

Thus, $\sum_{i=1}^I \tilde{x}_i^\epsilon \in V$

- As $\epsilon \rightarrow 0, \forall i, \tilde{x}_i^\epsilon \rightarrow x_i^*$ and therefore $\sum_{i=1}^I \tilde{x}_i^\epsilon \rightarrow \sum_{i=1}^I x_i^*$. Thus, $\sum_{i=1}^I x_i^* \in \overline{V}$.
- $\sum_{i=1}^I x_i^* = \bar{\omega} + \sum_{j=1}^J y_j^* \in \bar{\omega} + Y$.
- Therefore, $\sum_{i=1}^I x_i^* \in \overline{V} \cap (\bar{\omega} + Y)$ and the claim follows.



Claim v. (1) holds, that is to say $(\forall j, j = 1, \dots, J)$,

$$[p^* \cdot y_j^* \geq p^* \cdot y_j, \forall y_j \in Y_j] \text{ (profit maximization)}$$

Proof. Fix a firm j .

- $\forall y_j \in Y_j, y_j + \sum_{h=1, h \neq j}^J y_h^* \in Y$.
- By claim iv,

$$p^* \cdot (\bar{\omega} + y_j^* + \sum_{h=1, h \neq j}^J y_h^*) \geq p^* \cdot (\bar{\omega} + y_j + \sum_{h=1, h \neq j}^J y_h^*)$$

- Subtract the first and third terms to both sides to establish the claim.

Claim vi. (2) holds, that is to say

$$\text{for } i = 1, \dots, I, [x_i \succsim_i x_i^* \implies p^* \cdot x_i \geq p^* \cdot x_i^*]$$

Proof. Fix a consumer i .

- By claim iv, $p^* \cdot (x_i + \sum_{h=i, h \neq i}^I x_h^*) \geq p^* \cdot (x_i^* + \sum_{h=i, h \neq i}^I x_h^*)$
- Subtracting the second term to both sides, the claim and the theorem are established.

QED

Arrow's exceptional economy

A corollary to the WSWT

Corollary 1 (WSWC) Let $\mathcal{E}$ be an Arrow-Debreu economy that satisfies the hypothesis of [Theorem 2](#) and p^* be the price identified in that theorem. If, in addition, preferences are continuous, consumption sets are convex and for $i = 1, \dots, I$ there is a bundle $x'_i \in X_i$ such that $p^* \cdot x'_i < p^* \cdot x_i^*$, then there exists an income transfer $t \in \mathbb{R}^I$, $\sum_{i=1}^I t_i = 0$ such that (x^*, y^*, p^*, t) is a WE with transfers of $\mathcal{E}$.

- Unsatisfactory version of SWT; it uses assumptions on endogenous elements of the model.

Proof [sketch.]

Theorem 2 (1) and (3) establish that firms maximize profit at the proposed production plan and that aggregate demand equals aggregate supply.

It remains to show preference maximization, that $\forall i$

$$\forall i, x_i^* \in D_i(p^*, y^*, t_i).$$

Theorem 2 (2) states that

$$\forall i, x_i \succsim_i x_i^* \implies p^* \cdot x_i \geq p^* \cdot x_i^*$$

Claim. $\forall i, x_i \succ_i x_i^* \implies p^* \cdot x_i > p^* \cdot x_i^*.$

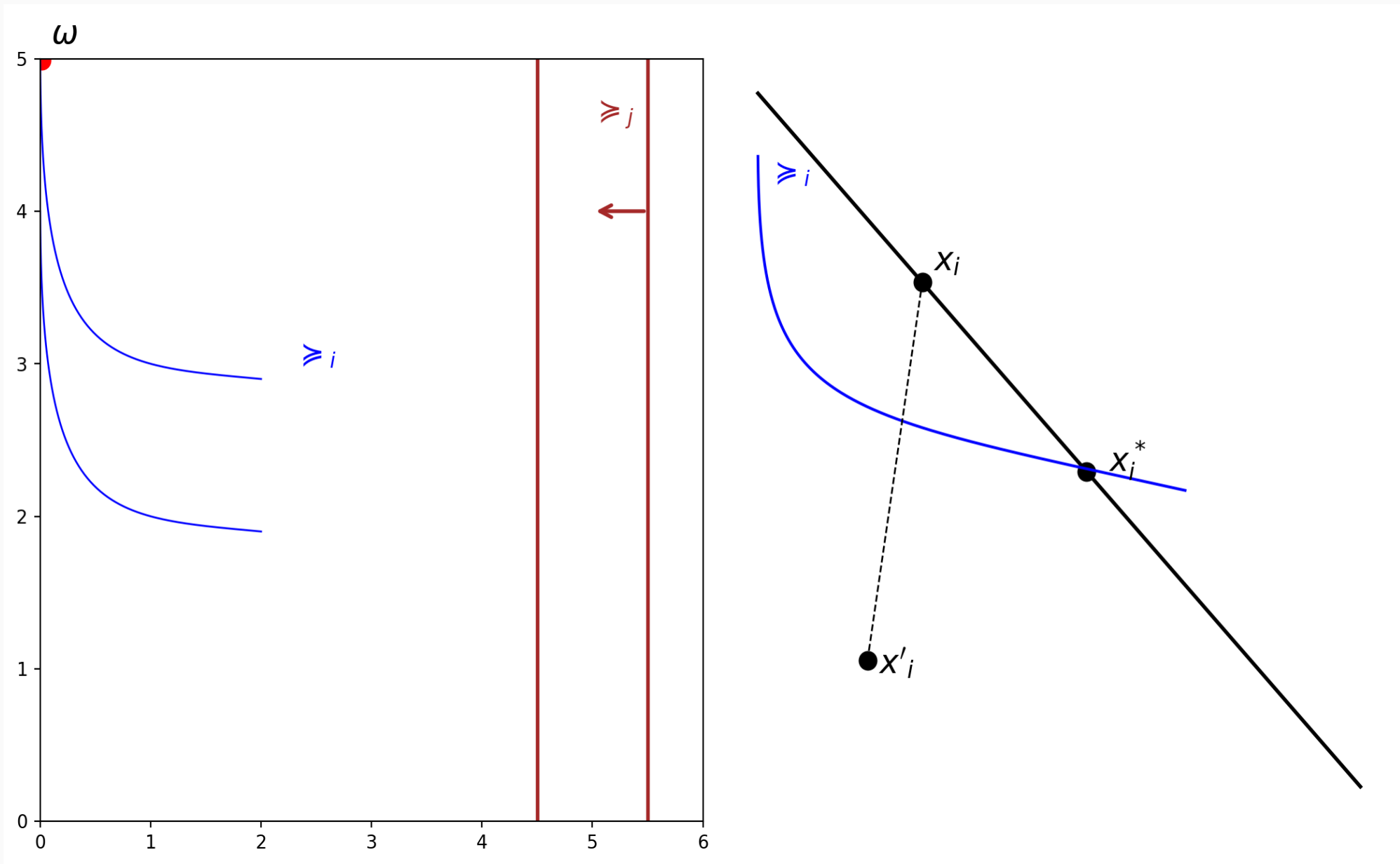
- Arguing by contradiction, suppose $\exists x_i \succ_i x_i^* \wedge p^* \cdot x_i = p^* \cdot x_i^*$
(using **Theorem 2** (2))
- Pick x'_i as in the hypothesis of corollary (x'_i costs less than x_i^*).
- By continuity of preferences

$$\exists \bar{\alpha} \in (0, 1) : 1 \geq \alpha \geq \bar{\alpha} \implies \alpha x_i + (1 - \alpha)x'_i \succ_i x_i^*$$

- Since $p^* \cdot x'_i < p^* \cdot x_i^* = p^* \cdot x_i$, then

$$\forall \alpha \in [0, 1), p^* \cdot [\alpha x_i + (1 - \alpha)x'_i] < p^* \cdot x_i^*$$

This is a contradiction of **Theorem 2** (2).



It remains to identify transfers $\{t_i\}_{i=1}^I$, $\sum_{i=1}^I t_i = 0$, so that $x_i^* \in D_i(p^*, y^*, t_i)$.

For $i = 1, \dots, I$, define,

$$t_i = p^* \cdot x_i^* - p^* \cdot \omega_i - \sum_{j=1}^J \theta_{ij} p^* y_j^*$$

Then

- $\sum_{i=1}^I t_i = 0$, so $t = (t_1, \dots, t_I)$ is an income transfer.
- By construction, $x_i^* \in B_i(p^*, y^*, t_i)$. By previous argument any $x_i, x_i \succsim_i x_i^*$ must cost more. Then $x_i^* \in D_i(p^*, y^*, t_i)$.

QED.

A version of the Second Welfare Theorem

Theorem 3 If x^* is a Pareto optimal allocation in a pure exchange economy, with strongly monotone, continuous, convex preferences, consumption sets are $\mathbb{R}_+^L$, and $\bar{\omega} \gg 0$, then there exists a price vector p^* and an income transfer t such that (x^*, p^*) is a Walrasian Equilibrium with transfers provided that $(\forall i) x_i^* \neq 0$.

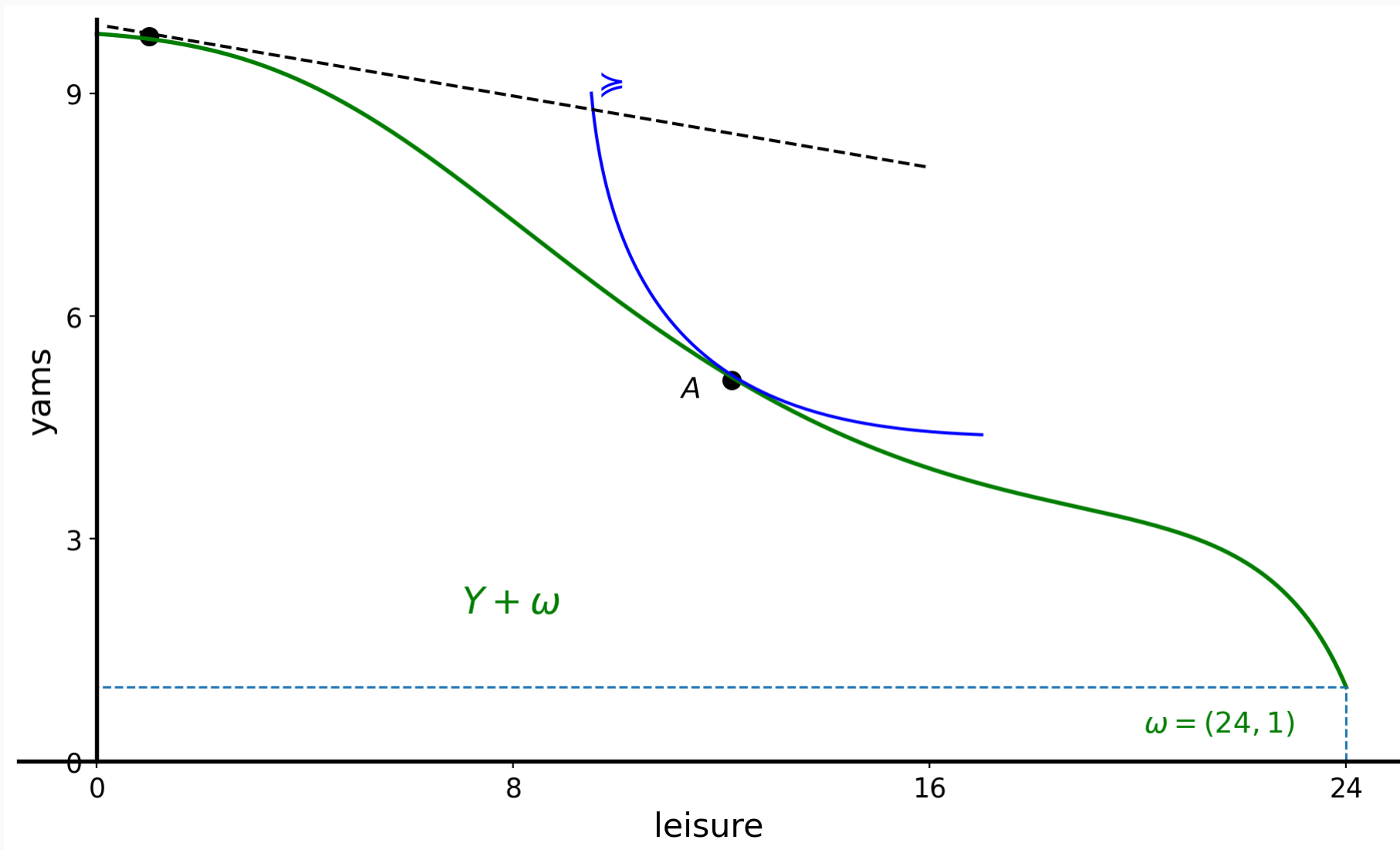
Failures of the SWT

- There are three general types of failures.
 - Failure for insufficient income.
 - Failure of LNS.
 - Failure of convexity, no appropriate prices exist.

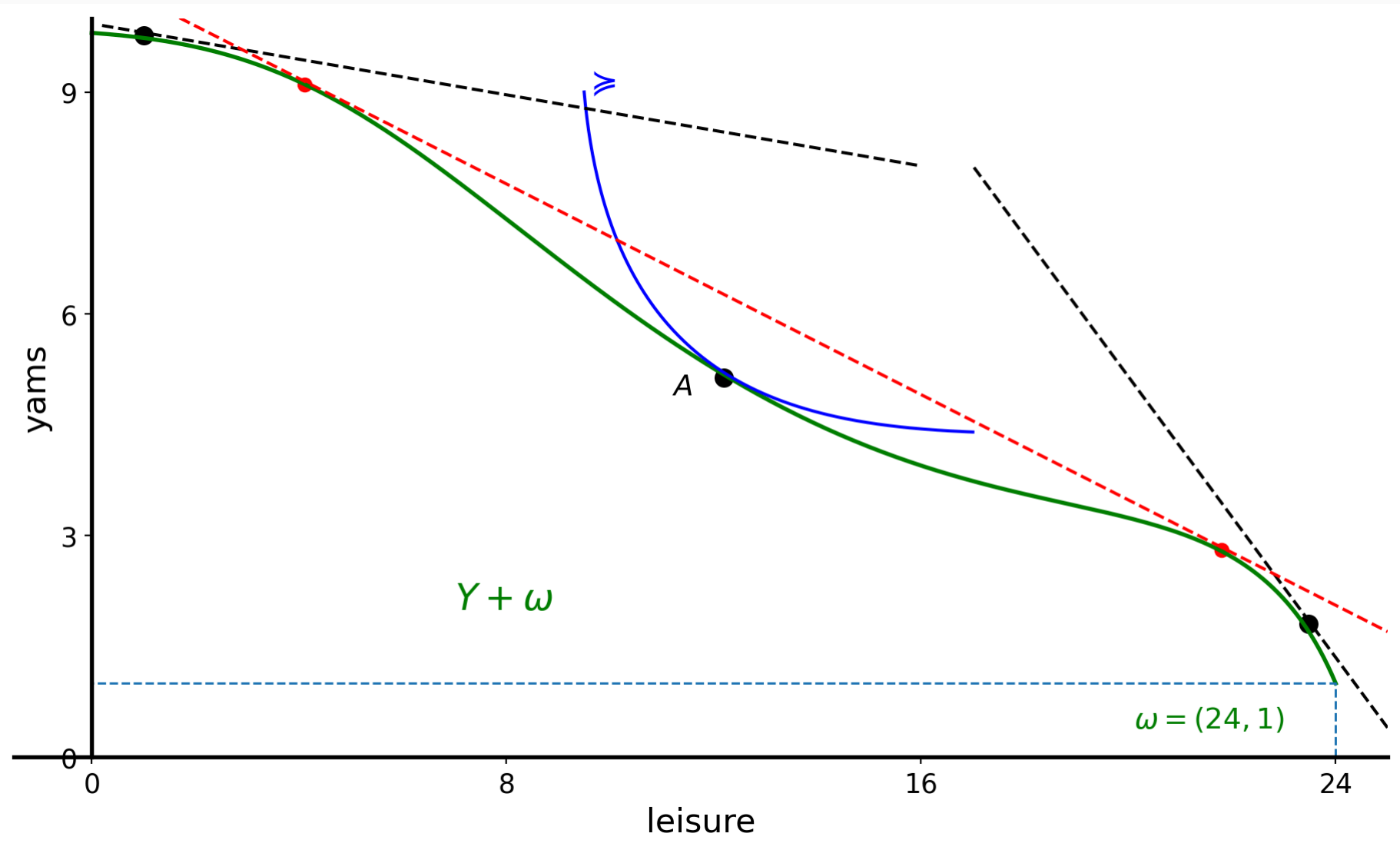
Failure of LNS

Failure of convexity

Failure of convexity in production



Failure of convexity in production



Interpretation and limitations

- Planner must ensure that the new supporting prices are taken as given. This fails with market power; outcome of market economy is not a WE.
- Planner's informational requirements: Identify the PO allocation to be implemented and compute the supporting prices in case transfers are used.
- In both points above, planner must know the statistical distribution of preferences and endowments.
- To implement transfers, the planner must observe the agents' private information, endowment and preferences. This is difficult; it often prevents lump sum transfers.

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- When lump-sum transfers are not possible, wealth redistribution is distortionary. Thus, the end result is not the desired PO allocation. This leads to theory of second best.
 - Transfers must be enforced.

Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. New York, Oxford: Oxford University Press.